Industrial Internship Report on

”Quiz Game”

Prepared by

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Executive Summary
This report presents the work completed during my Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. During this internship, I developed a Python-based Quiz Game that allows users to answer multiple-choice questions and view their score at the end of the quiz. The project helped me understand Python programming concepts such as variables, loops, conditional statements, functions, lists, dictionaries, and user input. It also improved my logical thinking and problem-solving skills through practical implementation. Working on this project gave me valuable hands-on experience in software development and taught me how to design, test, and document a real project. Overall, this internship was a great learning experience and strengthened my confidence in programming.

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1 Preface

This internship gave me an opportunity to improve my programming skills by working on a practical Python project. During the internship, I learned different Python concepts such as variables, loops, conditional statements, functions, lists, dictionaries, and user input. I applied these concepts while developing a Quiz Game application.

The project helped me understand how to solve problems step by step and how to organize code in a better way. I also learned the importance of testing a program and making it user-friendly.

I would like to thank **Upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd.** for providing this learning opportunity. I am also thankful to the mentors who guided me throughout the internship.

Overall, this internship has increased my confidence in Python programming and motivated me to continue learning new technologies and build more projects in the future.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

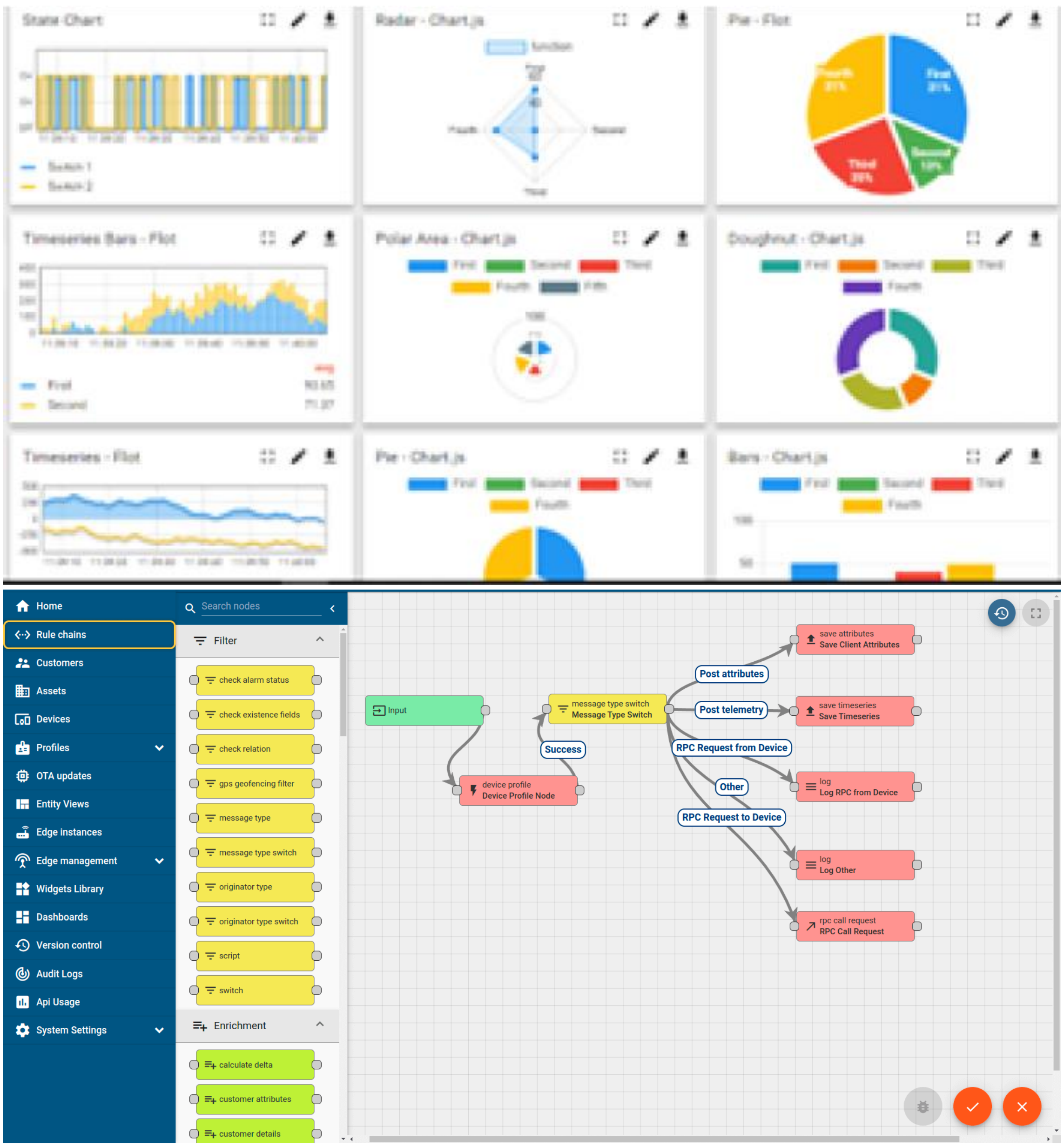
• Build Your own dashboard

• Analytics and Reporting

• Alert and Notification

• Integration with third party application(Power BI, SAP, ERP)

• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

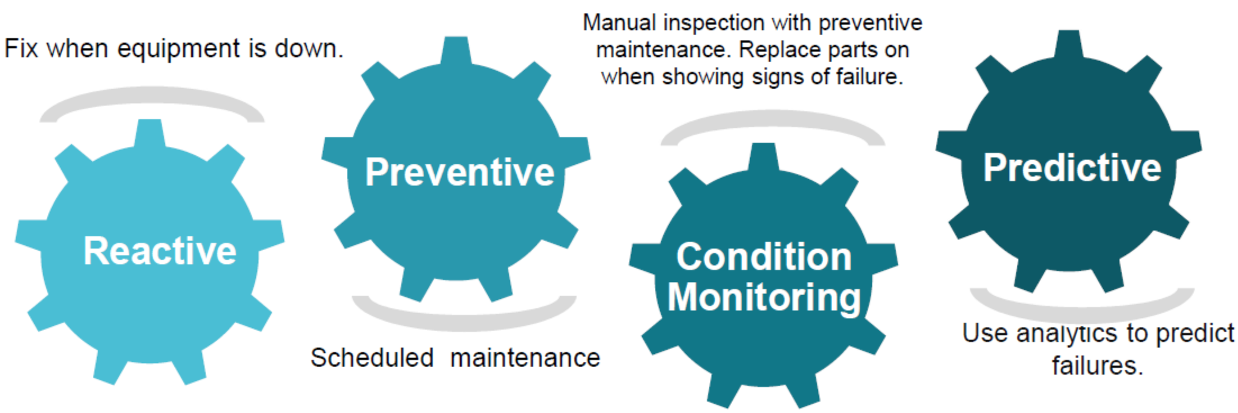
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

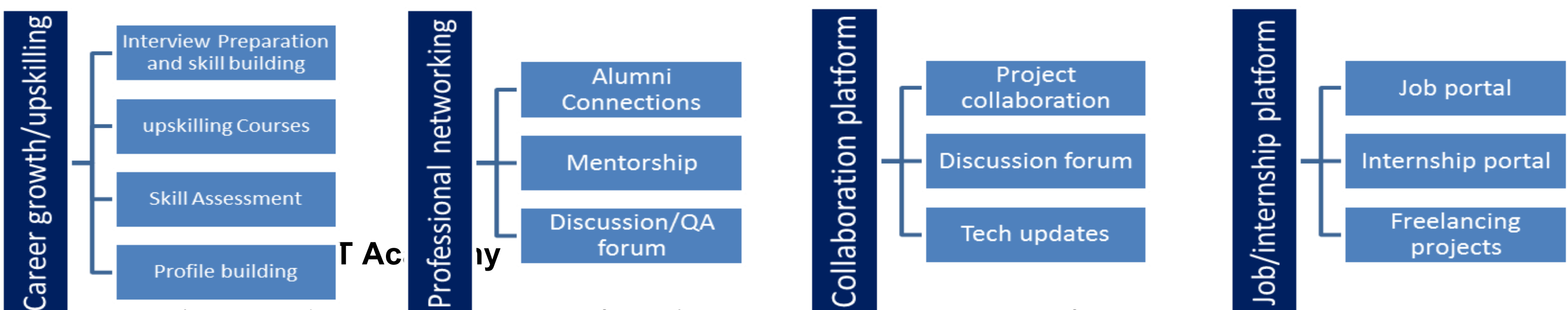
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



T Ac ny

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Official Documentation – https://docs.python.org/3/

[2] GitHub – https://github.com/

[3] Visual Studio Code Documentation – https://code.visualstudio.com/docs

2.6 Glossary

Terms	Acronym
Python	PY
Integrated Development Environment	IDE
Visual Studio Code	VS Code
GitHub	GH
Multiple Choice Question	MCQ

3 Problem Statement

The objective of this project is to develop a **Python-based Quiz Game** that allows users to answer multiple-choice questions and receive their results instantly. In many traditional quiz systems, questions are evaluated manually, which can be time-consuming and prone to errors. The proposed Quiz Game automates this process by checking answers, calculating scores, displaying percentages, assigning grades, and providing immediate feedback to the user.

The application is developed using Python and makes use of fundamental programming concepts such as variables, loops, conditional statements, lists, dictionaries, and input validation. The project aims to provide a simple, interactive, and user-friendly way to assess basic Python knowledge while improving programming and logical thinking skills.

4 Existing and Proposed solution

Existing Solution

Traditional quiz systems are often conducted using paper-based tests or simple online forms where evaluation may require manual effort. These methods can be time-consuming, less interactive, and may not provide instant feedback to the user. In many cases, users do not receive immediate scores or performance analysis.

Proposed Solution

The proposed solution is a **Python-based Quiz Game** that provides an interactive way to assess a user's knowledge through multiple-choice questions. The application checks answers automatically, calculates the total score, percentage, grade, and displays the final result instantly. It also validates user input and provides feedback after every question, making the quiz simple and user-friendly.

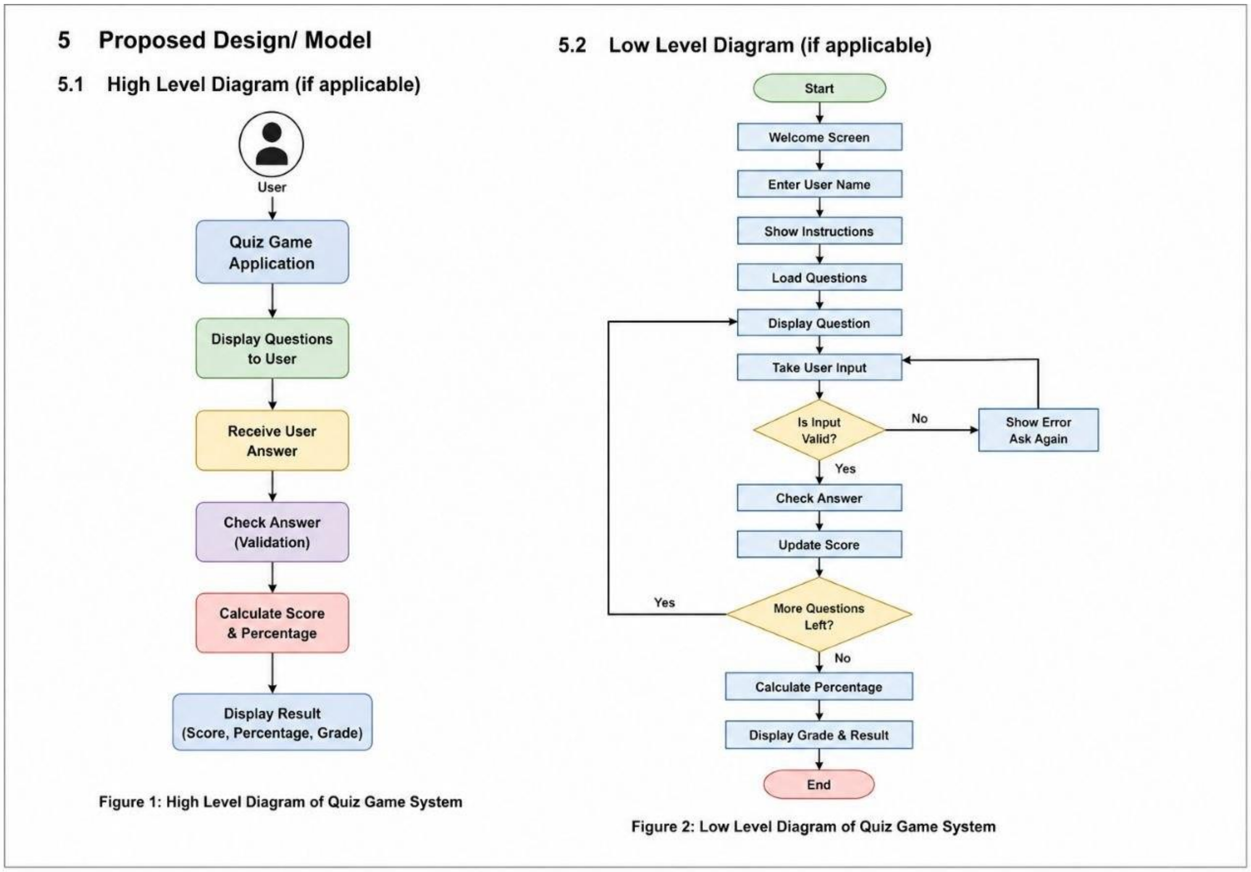
Value Addition

- Reduces manual evaluation.
- Provides instant results.
- Easy to use through the command line.
- Helps users practice Python concepts.
- Can be extended with a database, timer, GUI, and larger question bank in the future.

4.1 Code submission (Github link): https://github.com/yourusername/upskillCampus

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model



5.3 Interfaces

The flowchart below shows the overall working of the Quiz Game developed in Python. The program starts by displaying a welcome screen and asking the user to enter their name. After showing the instructions, the quiz begins by displaying one question at a time. The user's answer is checked for validity before comparing it with the correct answer. The score is updated after every question. Once all questions are completed, the program calculates the percentage, assigns a grade, and displays the final result. Finally, the user can choose whether to play the quiz again or exit the application.

6 Performance Test

The Quiz Game was tested on different user inputs to verify its correctness and performance. The application was checked for input validation, score calculation, result generation, and overall program flow. Since this is a console-based Python application, the memory usage and execution time are very low. The program performs efficiently for the given number of questions and provides instant feedback to the user.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Start the quiz	Welcome screen is displayed	Passed
Enter valid name	Quiz starts successfully	Passed
Enter invalid option	Error message is displayed	Passed
Enter correct answer	Score increases by 1	Passed
Enter wrong answer	Correct answer is shown	Passed
Complete quiz	Final score and grade displayed	Passed
Choose Play Again	Program asks to restart	Passed

6.2 Test Procedure

1. Launch the Python program.
2. Enter the user's name.
3. Read the instructions.
4. Answer all quiz questions.
5. Test both valid and invalid inputs.
6. Verify score, percentage, grade, and final result.
7. Test the Play Again option.

6.3 Performance Outcome

- Program executed successfully without errors.
- Input validation worked correctly.
- Score and percentage were calculated accurately.
- Grade and remarks were displayed correctly.
- The application responded instantly with minimal memory usage.
- Overall performance was satisfactory for a console-based quiz application.

7 My learnings

During this internship, I learned how to develop a complete Python project from planning to implementation. While building the Quiz Game, I improved my understanding of Python basics such as variables, loops, conditional statements, functions, lists, dictionaries, and user input.
I also learned how to organize code properly, validate user input, calculate scores and percentages, and display meaningful results. Working on this project helped me improve my logical thinking, debugging skills, and confidence in solving programming problems.
Apart from coding, I learned the importance of documentation and version control by creating a GitHub repository and preparing the project report. This internship gave me practical experience and motivated me to continue learning Python, Data Structures, and software development. I believe these skills will help me in future internships, placements, and software development projects.

8 Future work scope

Although the current version of the Quiz Game works successfully, there are several improvements that can be made in the future.

- Add multiple quiz categories such as Python, Java, C++, and General Knowledge.
- Store users' scores and quiz history in a database.
- Develop a graphical user interface (GUI) using Tkinter or PyQt.
- Introduce difficulty levels (Easy, Medium, Hard).
- Add a timer for each question.
- Randomize questions and options every time the quiz starts.
- Allow questions to be loaded from external files or a database.
- Generate a detailed performance report with weak and strong areas.
- Create an online version where multiple users can participate.

These enhancements would make the application more interactive, user-friendly, and suitable for real educational and training environments.